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Walter Wittke
Martin Wittke
Claus Erichsen
Bettina Wittke-Schmitt
Patricia Wittke-Gattermann
Dieter Schmitt

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AJRM as basis for design and construction of more than 70 km of tunnels of the Railway Project Stuttgart-Ulm

The tunnels of the railway project Stuttgart-Ulm are located in sedimentary rock of the Keuper Formation, the Black Jurassic, the Brown Jurassic and the White Jurassic formation. The two longest tunnels are excavated by tunnel boring machine over most of their length. The remaining tunnels, which the authors are involved in, are excavated by conventional tunnelling. All tunnels are designed on the basis of the AJRM-method. This method is based on an anisotropic, elasto-viscoplastic model for jointed rock and corresponding FE-software, which has been extended to seepage and swelling. The method has been applied to practical projects for the last 40 years. By means of rock mechanical test programs and back-analyses of projects in a variety of rocks, the reliability of the predictions is quite high and the application leads to safe and economic solutions.
Special emphasis is given to tunnels located in swelling gypsum Keuper. The phenomena of swelling and leaching are described and a swelling model is outlined. The evaluation of the rock mechanical parameters and a method of analyses are explained. On this basis, the design principles and package of measures applied for the tunnels located in anhydrite-bearing rock are described. Some important aspects with regards to tunnelling in leached Gypsum Keuper and in the other Keuper layers as well as the Black Jurassic are addressed. For the 8.8 km long Boßler Tunnel, located in the Brown and White Jurassic, for an approx. 900 m long section in the Aalenium 2 with a maximum overburden of 280 m squeezing rock conditions had to be anticipated. By means of a vertical exploration shaft and a connecting exploration gallery, an improvement of the predicted characteristic rock mechanical parameters could be justified, thus enabling TBM-tunnelling also in this section, with an increase of the segment thickness from 45 to 65 cm. For the Steinbühltunnel located in White Jurassic, special emphasis was placed on karstic rocks. The corresponding measures are described.

1 Introduction

At present the central station of the city of Stuttgart is a dead-end station. The trains coming from Mannheim via the high-speed railway line to Stuttgart enter the city above ground. After change of direction of driving, the alignment for the trains towards Munich via the city of Ulm follows the Neckar valley and winds up the Swabian Alb and again downwards to Ulm at the eastern side of the mountain chain.

In the future the main station will be an underground through-going station. It will be connected to the high-speed line from Mannheim by the Feuerbach Tunnel (Figure 1).

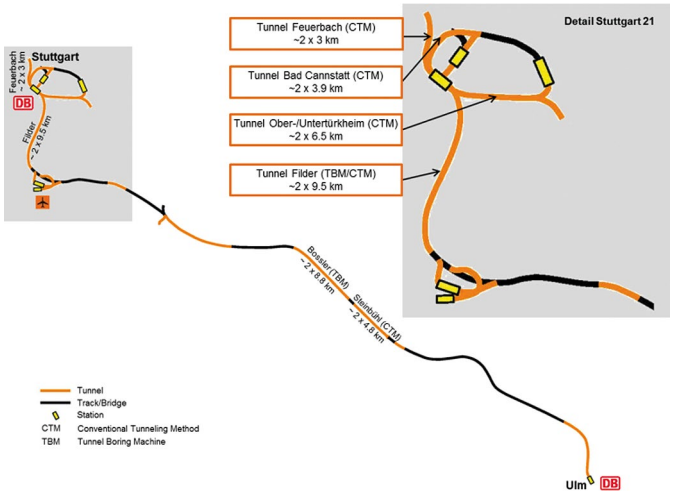


Fig. 1. Railway Project Stuttgart-Ulm

Since Stuttgart is located in a valley, the line goes uphill to the airport and the Filder plane through the 9.5 km long Filder Tunnel. From there the new high-speed line follows the existing highway A8 and after passing through the Albvorland Tunnel climbs up to the Swabian Alb through the 8.8 km long Boßler Tunnel and the 4.8 km long Steinbühl Tunnel. On the other side of the mountain chain, the Albabstieg Tunnel connects the new high-speed line to the existing station of the city of Ulm. Within the city of Stuttgart the tunnel tubes towards Bad Cannstatt and towards Ober-/Untertürkheim connect the new station to the existing lines and to a new maintenance station (see Figure 1) [1].

Except the Albvorland Tunnel and the Albabstieg Tunnel, the authors are involved in all of the tunnels as ground and tunnel experts. Furthermore, the design for execution for some sophisticated structures is being elaborated by WBI. Co-workers of WBI are also on-site to support the client.

Except certain sections, the two tubes of the Filder Tunnel and the Boßler Tunnel are being excavated by a TBM [2] [3]. All other tunnels are constructed by means of the conventional tunnelling method. In total, WBI is responsible for more than 70 km of tunnels of this project. The tunnels of the project pass through all layers of the Keuper-Formation. Furthermore, they are located in the Black Jurassic, the Brown Jurassic and the karstic White Jurassic formation (Figure 2) [4].

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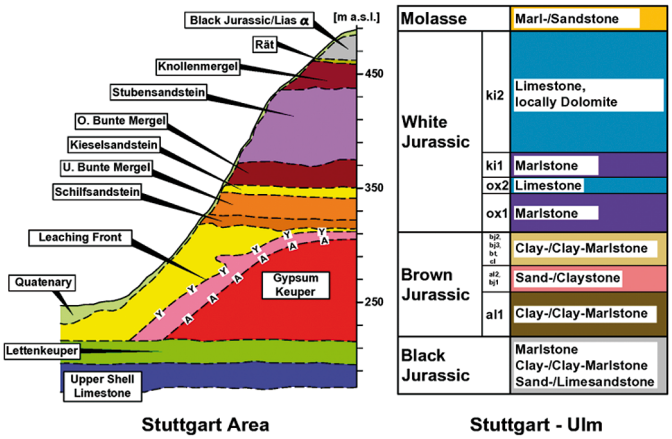


Fig. 2. Geological conditions for the tunnels of the project

The basis for the design of the tunnels in the above mentioned sedimentary rocks is the so-called AJRM-method [5]. For all layers different aspects of rock engineering have to be observed during design and construction. The present article is concentrated on the problems resulting from swelling Gypsum Keuper. Furthermore, a short description of the other layers of the Keuper formation and the Black Jurassic encountered by the Filder Tunnel will be given. For the Brown Jurassic, squeezing rock conditions were anticipated in the tender phase, whereas in the White Jurassic, karstification was the main issue to be considered.

2 Tunnels of the project Stuttgart 21

2.1 Gypsum Keuper with Anhydrite and Gypsum

More than 34 km of tunnels of the project are located in the Gypsum Keuper formation. The claystones of the so-called unleached Gypsum Keuper in the original state contain anhydrite $CaSO_4$ (Figure 3). If anhydrite comes into contact with water, it goes into solution and, if the solution is not drained off, crystallization occurs, resulting in Gypsum, which contains two molecules of water ($CaSO_4 \cdot 2H_20$):

$$CaSO_4 + 2H_2O \rightarrow CaSO_4 \cdot 2H_2O$$

This process is combined with a volume increase of 61 %. Since most of the layers of the Gypsum Keuper originally contain approx. 30 % of anhydrite, this process in-situ theoretically is connected with a volume increase of 18 % [5]. Under confined conditions this swelling leads to high swelling pressures. For full transformation of the available anhydrite and full confinement, max. pressures of up to 10 MN/m^2 must expected [5] [6]. To achieve full transformation of the sulfate in 1 m^3 of rock, approx. 230 l of water are required [7].

In spite of the very low permeability of the anhydrite-bearing rock, water has penetrated into the rock and initiated the transformation into gypsum in geological times down to a certain depth. Thus, adjacent to the surface of the anhydrite-bearing rock, rock in which the sulfate content consists of gypsum is existing [8]. On the surface of the gypsum rock, dissolution of gypsum takes place due to groundwater flow, leading to water channels and small karstic holes. This zone has a high permeability. Above

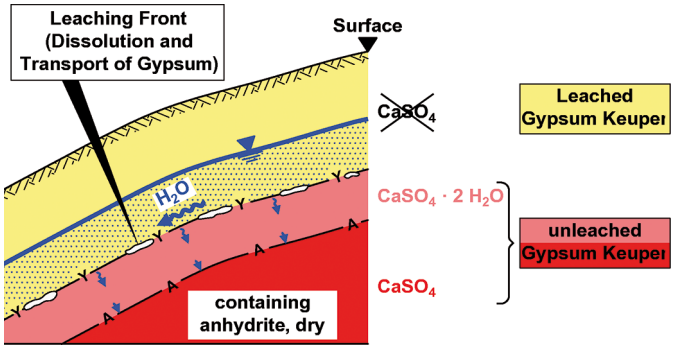
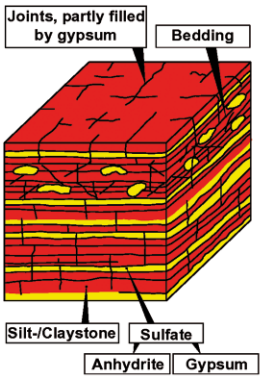


Fig. 3. Gypsum Keuper formation, principle sketch



		Order of Magnitude	Evaluation
Unit Weight	γ [kN/m³]	27	Testing
Elastic Behavior	E [MN/m²]	4000	Back Analyses Plate Load Test Dilatometer
	ν [-]	0,25	Unconfined compressive tests
Discontinuities (Bedding/Joints)	d [m]	0,05 / 1	Mapping Scanning in Boreholes Back Analyses
	c [MN/m²]	0 / 0,5	
	φ [°]	30 / 35	
	ψ [°]	10 / 10	
Permeability	k_f [m/s]	10^{-9}	Lugeon Tests Back Analyses
Water Uptake	D [m²/s]	$2 * 10^{-12}$	Testing Back Analyses
	S_D [-]	depending on content of anhydrite	Mineralogy
Swelling	K_q [-]	0,1 - 0,2	Tests Back Analyses
	σ_{q0} (ω = 1) [MN/m²]	9 - 10	

Fig. 4. Unleached Gypsum Keuper, rock mechanical model

this so-called leaching front, the sulfates are dissolved and Gypsum Keuper is transferred to a residual rock with a permeability of a silt (see Figure 3).

The unleached rock is horizontally bedded (Figure 4). Only minor jointing exists, the joints being perpendicular to the bedding. Some discontinuities also reveal an inclined dip [5]. The parameters describing the deformability of the rock and the strength along the discontinuities have been evaluated from laboratory and field tests carried out for the on-going as well as for projects already executed. Also back-analyses have formed the basis for determination of the parameters [5]. Extrapolation from other projects in the same rock is possible because the Gypsum Keuper is rather homogeneous throughout the area of Southwest Germany.

The coefficient of permeability given in Figure 4 results from Lugeon-tests carried out for the Stuttgart project and

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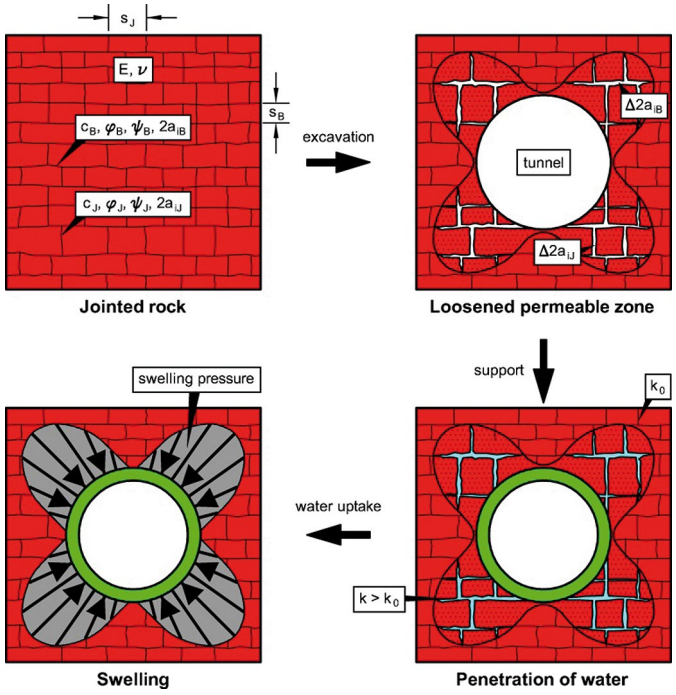


Fig. 5. Phenomena when tunnelling in swelling rock, principle

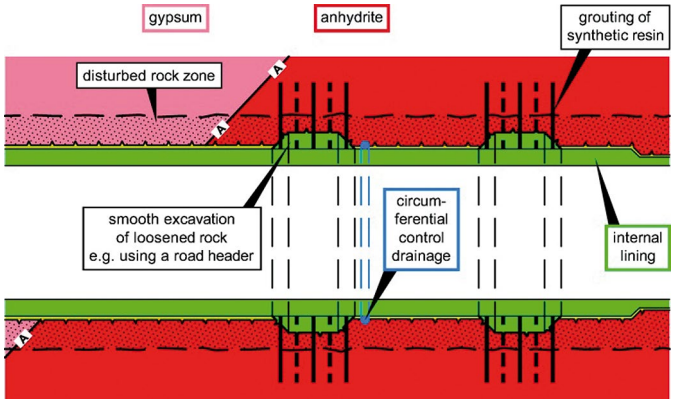


Fig. 6. Sealing structure

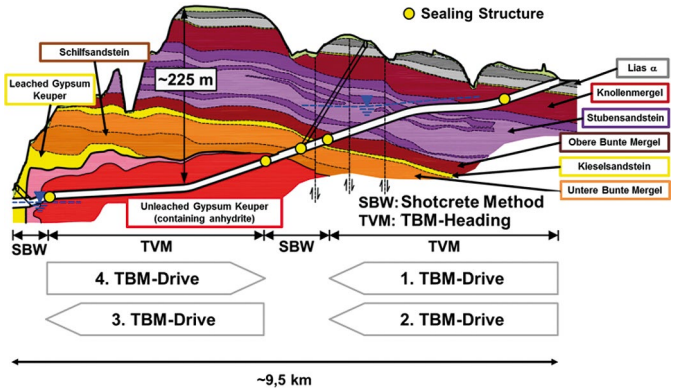


Fig. 7. Longitudinal section of the Filder Tunnel

from back-analyses carried out for a large-scale field test in a test gallery adjacent to the Freudenstein Tunnel [5] [9] [10] [11]. The given value is valid for the horizontal bedding. In vertical direction, due to the limited jointing, the permeability can be expected to be somewhat smaller.

According to the model presented in [10], the water uptake of the intact rock and the time for transformation from anhydrite into gypsum are described by the Diffusion Model. The two required parameters were evaluated by laboratory tests, back-analyses and determination of the sulfate content [5] [10] [11].

The relationship between swelling stress and strain is described by an extended Huder-Amberg-Model [10] and requires the two parameters given in Figure 4. These are derived from laboratory tests and back-analyses [6].

The phenomena to be considered within the modelling for tunnels in swelling rock are illustrated in Figure 5 [10]. Starting from the undisturbed condition (Figure 5, top left), excavation of the tunnel leads to a stress re-distribution around the opening which in-turn leads to dilatant displacements due to shear along the discontinuities and thus to an increase of the permeability (Figure 5, top right) [5]. If this zone of higher permeability has contact to water-bearing layers, water can penetrate due to seepage towards the rock adjacent to the tunnel (Figure 5, bottom right). As a consequence, swelling is initiated and leads to a modification of the stress field and the permeability adjacent to the tunnel (Figure 5, bottom left). The described process is implemented in a 3D FE-program and is simulated in iterative analyses [10].

The described model thus includes an elasto-visco-plastic stress-strain-law, transient laminar seepage flow, water consumption/capillary suction by diffusion and a swelling law, which are fully coupled and enable consideration of anisotropic jointed rock [10]. All mechanical and hydraulic parameters of the in-situ rock mass are taken as input parameters. The construction sequence and e.g. 100 years of swelling are simulated. The analyses lead to stresses and displacements in the system and to the stress resultants of the concrete lining. The change of permeability as a consequence of construction and swelling results from the analyses.

It is obvious that the swelling process only can occur if water has access to the anhydrite-bearing rock. In order to avoid or minimize the access of water, a number of measures are foreseen for the project Stuttgart 21.

In order to interrupt seepage flow from water-bearing strata through the loosened zone in the rock immediately adjacent to the tunnel, so-called sealing structures are being constructed. They consist of two approx. 1 m thick and 4 to 5 m long concrete rings (Figure 6). These, because of their limited length, lead to only minor dilation in the rock. During optimization of the project it was decided that these rings, which need to be water-tight throughout the lifetime of the tunnels, will be constructed by steel fibre-reinforced shotcrete. Sealing of the adjacent rock is carried out by chemical grouting through radial drillholes.

For the Filder Tunnel, three of these sealing structures are planned at the upper edge of the anhydrite-bearing zone and one sealing structure will be constructed near the new central station in the valley (Figure 7). Since these sealing structures cannot be constructed early enough from the TBM-driven tunnel, TBM tunnelling is limited to the upper and the lower part of the Filder Tunnel as given in Figure 7.

From the water-bearing layers above the anhydrite-bearing rock in the Filder Tunnel, no water can reach the

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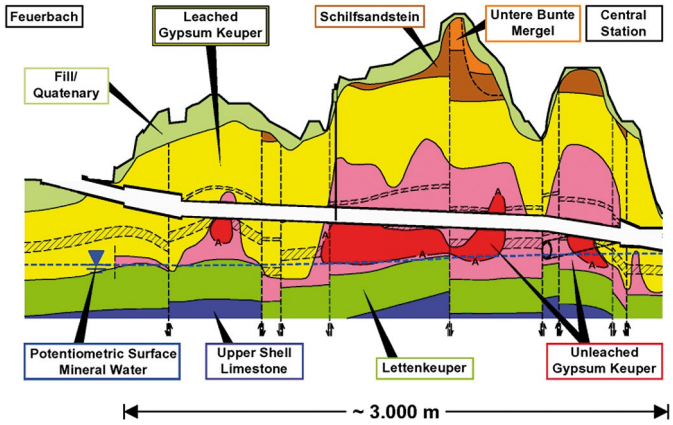


Fig. 8. Longitudinal section of the tunnel Feuerbach

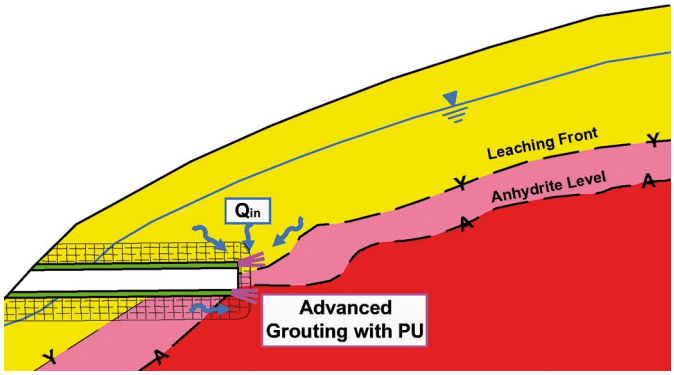


Fig. 9. Advanced grouting in the area of the leaching front, principle

tunnel, because the loosened zone around the tunnel here does not reach to the upper boundary of the sulfate rocks (see Figure 7). This is at least locally different for the tunnels to Feuerbach and Bad Cannstatt (Figure 8). Here it is important to seal the highly permeable zone of the leaching front (see Figure 3) by advanced grouting (Figure 9). This grouting has been successfully carried out by Polyurethane, which enabled practically dry tunnelling through the leaching front. Furthermore, the more permeable loosened zone around the tunnel (see Figure 5, top right) will be grouted after completion of excavation and support by a low-viscous chemical grout (Figure 10) [12].

For areas where the leaching front is located well above the tunnel a circular tunnel is being excavated. In cases when the anhydrite level is located in the lower part of the cross-section and water-bearing strata are close to or located in the area of the cross-section, a U-profile has proven to be adequate. By means of this, water penetration from above is minimized [13]. Results of a corresponding grouting test field will be presented at the next "Felsmechanik- und Tunnelbau-Tag" in May 2017 in Weinheim.

In summary, the following package of measures is applied for tunnelling in swelling rocks:

WBI
Prof. Dr.-Ing. W. Wittke Beratende Ingenieure für Grundbau und Felsbau GmbH
Worldwide Engineering
Im Technologiepark 3 69469 Weinheim wbi@wbionline.de www.wbionline.de

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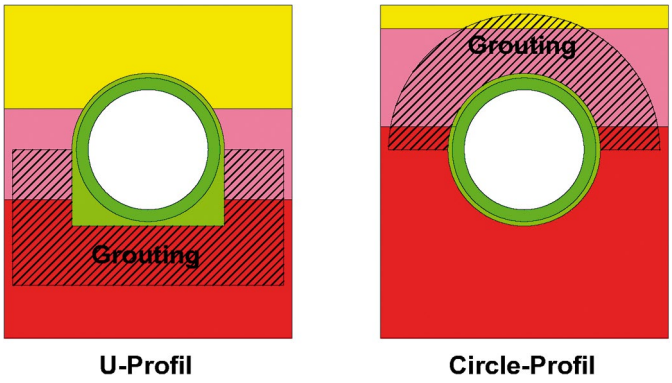


Fig. 10. U-profile and circular profile, principle of grouting

- Several decades of research and practice:
  - Modelling
  - 3D numerical analyses
  - Calibration and validation by means of laboratory and field tests as well as on the basis of tunnels executed in the past
- Selection of cross section (circle and U-profile respectively):
  - With the aim to reduce and minimize respectively the access of water
  - With regards to stability
- Excavation and support:
  - Dry tunnelling
  - Early closure of invert
  - Stiff shotcrete support
  - No anchors
- Sealing:
  - Advanced grouting to seal the water-bearing leaching front
  - Post-grouting to seal discontinuities in rock
  - Construction of sealing structures to avoid longitudinal groundwater flow in the rock along the tunnel
- Monitoring and observation during and after construction:
  - Evaluation of sulfate content by sampling from the face and from radial drillings
  - Monitoring of displacements during construction
  - Long-term monitoring of displacements and stresses of the tunnel lining and in the adjacent rock

According to the knowledge of the authors, some of these measures, such as the model and the corresponding analyses as well as grouting of acrylate gel, and especially the combination of measures are applied for the first time in a tunnel in swelling rock.

2.2 Leached Gypsum Keuper, residual rock

A major part of the tunnels to be constructed within the city of Stuttgart are located in leached Gypsum Keuper (see Figures 3 and 8). Because of the high deformability and the comparatively low strength of this residual rock, special attention has to be given to the stability during construction and to the subsidence at the surface [4] [14].

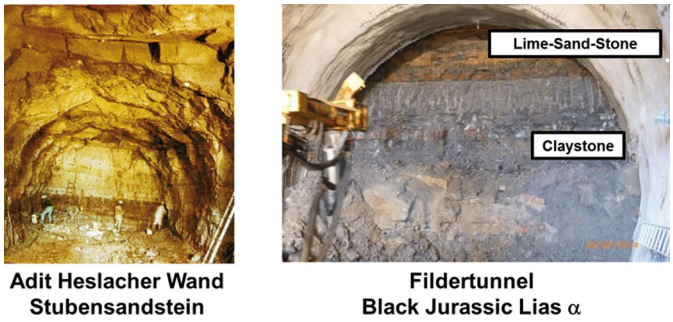


Fig. 11. Stubensandstein and Lias α

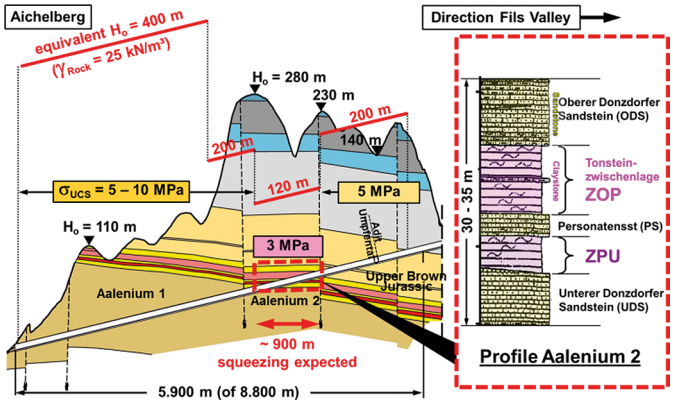


Fig. 12. Western section of Boßler Tunnel, longitudinal section and unconfined compressive strength

2.3 Marlstones, Sandstones and Claystones

The upper part of the Filder Tunnel is located in marlstones, sandstones and claystones (Bunter Mergel, Kieselsandstein, Stubenstandstein and Black Jurassic), which are more or less horizontally bedded and vertically jointed (see Figure 7), such as shown on the left hand side of Figure 11 for the Stubensandstein (Keuper sandstone) and for limestone and claystones of the black Jurassic formation (Figure 11, right).

The Kieselsandstone, the Stubensandstein and locally also the sandstones in the Black Jurassic reveal a comparatively high permeability. In the Stubendsandstein a groundwater pressure of up to 6 bar has to be accounted for.

Of special concern is the clogging potential of the claystone layers of the Stubensandstein. Because of the comparatively high natural water content, the claystone layers of the Stubensandstein are prone to clogging. A certain inflow of water during excavation leads to a decrease of the consistency index from solid to semi-solid and stiff.

Another issue is the subsidence in the section located in the Black Jurassic formation, which is located immediately underneath sensitive buildings. Here the excavation contour in claystones was stabilized by compressed air and practically zero settlements could be achieved.

3 Ascent to the Swabian Alb of the highspeed railway line Wendlingen-Ulm

3.1 Brown Jurassic

The geological longitudinal section of the first 5.9 km long section of the 8.8 km long Boßler Tunnel is represented in

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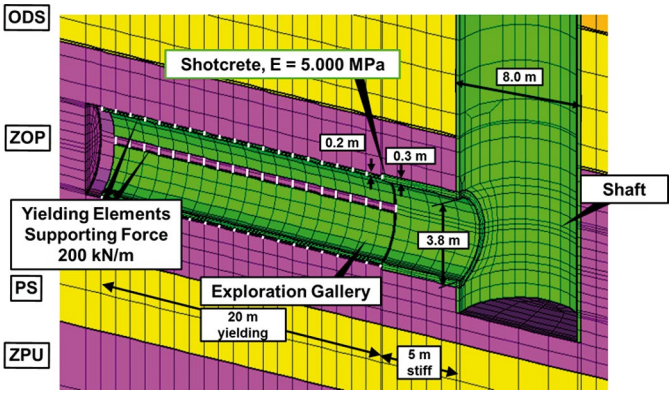


Fig. 13. Exploration shaft and gallery, 3D FE-mesh

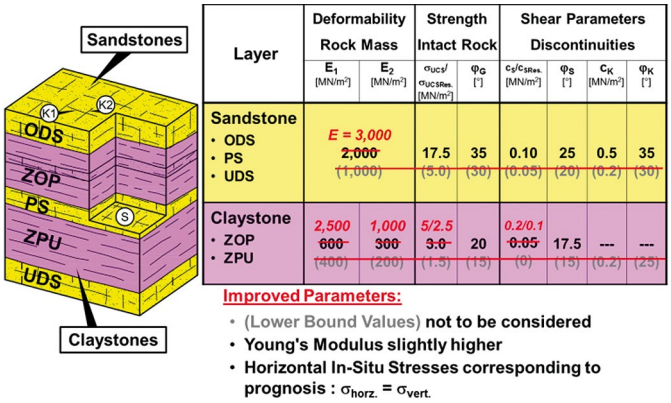


Fig. 15. Aalenium 2, rock mechanical model

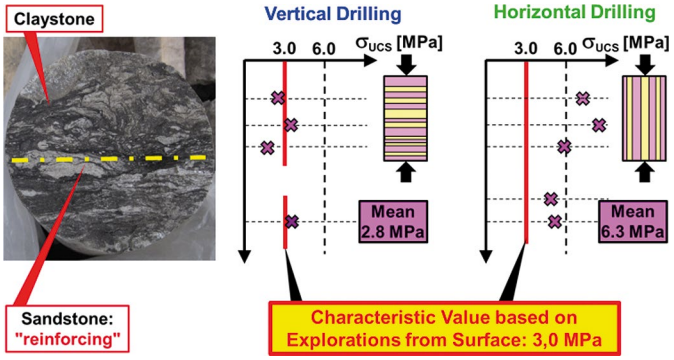


Fig. 14. Claystone of ZOP and unconfined compressive strength

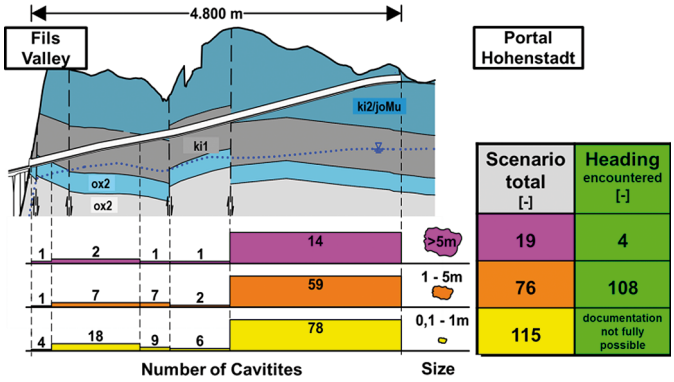


Fig. 16. Steinbühltunnel, longitudinal section and karstification

Figure 12. The west section is located in the different layers of the Brown Jurassic formation. The eastern section of the tunnel, which is only partly represented in Figure 12, is located in White Jurassic.

The section in Aalenium 2, with a maximum overburden of 280 m, was considered critical with regards to squeezing rock from the beginning [15]. The unconfined compressive strength of the claystone layers within this formation (ZPU and ZOP) was determined on samples taken from the exploratory boreholes to only 3 MN/m^2. This unconfined compressive strength corresponds to an overburden pressure resulting from an overburden of 120 m (see Figure 12). Thus, it is much smaller than the stresses resulting from the max. overburden of 280 m. Consequently, in the tender, it was foreseen to excavate the corresponding section conventionally with a yielding support.

After the contract was awarded, it was discussed whether an exploration of the layer in question by means of a gallery might lead to a higher strength. This would have enabled to excavate the whole tunnel by a TBM, as it was permitted in the tender for all other layers except the Karstic White Jurassic [3] [15] [16].

Thus, after the intermediate access Umpfental had been excavated, it was decided to excavate a vertical shaft at the intersection of adit with the main tunnel. This vertical shaft extended down to the ZOP-layer of the Aalenium 2, in which a horizontal gallery with a yielding support was excavated (Figure 13).

The shaft and the gallery were intensely geotechnically mapped and monitored during construction. In addition, core drillings were carried out to explore the corresponding rock mass and to take samples from ZOP and ZPU layers (Figure 14). A close view of horizontal cores, which also were taken, shows that sandstone lenses are included in the claystone of the ZOP and ZPU layers (Figure 14, left). Laboratory testing has shown that these sandstone intercalations lead to a higher unconfined compressive strength for loading parallel to bedding (Figure 14, right), whereas for loading perpendicular to bedding, practically the same strength as predicted was measured (Figure 14, centre). Also the Young's modulus of the intact rock was higher than anticipated.



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These test results and back-analyses of the displacements measured in the shaft and gallery led to a modification of the rock mechanical parameters in comparison to the prediction made in the tender phase [15]. The revised parameters are listed in Figure 15 and compared to those given in the tender. From this it could be concluded that the lower-bound parameters given in the tender could be eliminated and that the characteristic parameters were slightly improved with regards to deformability.

On this basis, it was possible to design and construct the Boßler Tunnel also in the section located in Aalenium 2 by TBM with a thickness of the segments increased from 45 to 65 cm [3] [15] [16].

It should be noted that the authors recommended to carry out a test gallery already during the exploration phase in order to investigate the issue described above.

3.2 White Jurassic

The eastern part of the Bosslertunnel and the 4.8 km long Steinbühltunnel are located in lime- and marlstones of the white Jurassic. Especially the eastern part of the Steinbühl Tunnel, located in the Kimmeridge 1 formation, is prone to karstification (Figure 16).

The Steinbühl Tunnel was excavated by means of conventional tunnelling. In order identify potential karstic holes in advance of the excavation, percussion drillings were carried out systematically with monitoring of the rate of advance, the torque, the pressure to be applied and the rounds per minute (MWD). The data were successfully interpreted so that larger karstic holes could be identified prior to excavation.

During the tender phase a scenario for occurrence of karstic openings to be expected during tunnel driving was predicted. This was done on the basis of the drillholes and the overall experience on the karstification in the area. The comparison between the predicted and the encountered number of openings reveals a reasonable good agreement (see Figure 16).

The surrounding of the tunnel after excavation was also explored by drillholes and cross-hole measurements by georadar in order to identify larger karstic holes and their impact on the design of the internal lining.

4 Summary

The authors support the German railway as ground and tunnel experts as well as designer for special structures for more than 70 km of tunnels of the railway project Stuttgart-Ulm. The tunnels are located in sedimentary rock of the Keuper Formation as well as of the Black Jurassic, the Brown Jurassic and the White Jurassic formation. The two longest tunnels are excavated by a TBM over most of their length. The remaining tunnels are excavated by conventional tunnelling. All of the tunnels are designed on the basis of the AJRM-method. This method is based on an anisotropic, elasto-viscoplastic model for jointed rock and corresponding FE-software, which has been extended to seepage and swelling. The method has been applied to practical projects for the last 40 years. By means of rock mechanical test programs and back-analyses of projects in a variety of rocks, the reliability of the predictions is quite high and the application leads to safe and economic solutions.

In the given paper, special emphasis is given to the tunnels located in swelling Gypsum Keuper. The phenomena of swelling and leaching are described and a model for the swelling rock is outlined. The evaluation of the corresponding rock mechanical parameters is explained and a method of analyses is briefly explained. On this basis, the design principles and package of measures applied for the tunnels located in anhydrite-bearing rock are described in detail.

Some important aspects with regards to tunnelling in the leached Gypsum Keuper as well as in the sandstones, marlstones and claystones of the other Keuper layers and the Black Jurassic are addressed.

For the 8.8 km long Boßler Tunnel, which is located in the Brown and White Jurassic, for an approx. 900 m long section in the Aalenium 2 with a maximum overburden of 280 m squeezing rock conditions had to be anticipated. By means of a vertical exploration shaft and a connecting exploration gallery, a slight improvement of the predicted characteristic rock mechanical parameters could be justified, thus enabling TBM-tunnelling also in this section, with an increase of the segment thickness from 45 to 65 cm.

For the Steinbühl Tunnel located in karstic rocks of the White Jurassic, special emphasis had to be placed on karstic rocks. The corresponding measures are briefly described.

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Prof. Dr.-Ing. Walter Wittke
WBI GmbH Prof. Dr.-Ing. W. Wittke
Beratende Ingenieure für Grundbau und Felsbau
Im Technologiepark 3
69469 Weinheim
Germany
wbi@wbionline.de



Dr-Ing. Martin Wittke
WBI GmbH Prof. Dr.-Ing. W. Wittke
Beratende Ingenieure für Grundbau und Felsbau
Im Technologiepark 3
69469 Weinheim
Germany
wbi@wbionline.de



Dr.-Ing. Claus Erichsen
WBI GmbH Prof. Dr.-Ing. W. Wittke
Beratende Ingenieure für Grundbau und Felsbau
Im Technologiepark 3
69469 Weinheim
Germany
wbi@wbionline.de



Dr.-Ing. Bettina Wittke-Schmitt
WBI GmbH Prof. Dr.-Ing. W. Wittke
Beratende Ingenieure für Grundbau und Felsbau
Im Technologiepark 3
69469 Weinheim
Germany
wbi@wbionline.de



Dr.-Ing. Patricia Wittke-Gattermann
WBI GmbH Prof. Dr.-Ing. W. Wittke
Beratende Ingenieure für Grundbau und Felsbau
Im Technologiepark 3
69469 Weinheim
Germany
wbi@wbionline.de



Dipl.-Ing. Dieter Schmitt
WBI GmbH Prof. Dr.-Ing. W. Wittke
Beratende Ingenieure für Grundbau und Felsbau
Im Technologiepark 3
69469 Weinheim
Germany
wbi@wbionline.de

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